Diseases of Carrot

1. Cercospora leaf spot - *Cercospora carotae*, *C.apii*
2. Alternaria leaf blight - *Alternaria dauci*, *A. radicina*
3. Powdery mildew - *Erysiphe polygoni*, *E. heraclei*

Bacterial diseases

4.Bacterial blight - *Xanthomonas campestris* pv *carotae*
5. Root scab - *Streptomyces scabies*
6. Bacterial soft rot - *Erwinia carotovora* pv. *Carotovora*, *E. carotovora pv. Atroseptica*

Phytoplasma

7. Carrot Yellows /Aster yellows- Phytoplasma

Carrot - Cercospora leaf spot *Cercospora carotae, C.apii*

Tan, brown or almost black spots – pale necrotic center with diffused chlorotic border

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On petioles, stems- elliptical, brownish lesions with pale center

Carrot - Cercospora leaf spot *Cercospora carotae, C.apii*

On leaf - Initially small lesions **at the margins**

Carrot - Cercospora leaf spot *Cercospora carotae, C.apii*

Entire leaflet withers and dies

Carrot - Cercospora leaf spot *Cercospora carotae, C.apii*

Severe on young leaves

Loss in foliage - the carrots -left behind during mechanical harvesting

Does not attack the fleshy roots

Symptoms:

- On leaf -lesions - initially small, **at the margins-lateral curling**
- **Roughly circular to elongate**, chlorotic,
- **Tan, brown or almost black spots** - **pale necrotic center with diffused chlorotic border**

Carrot - Cercospora leaf spot *Cercospora carotae, C.apii*

Symptoms contd…

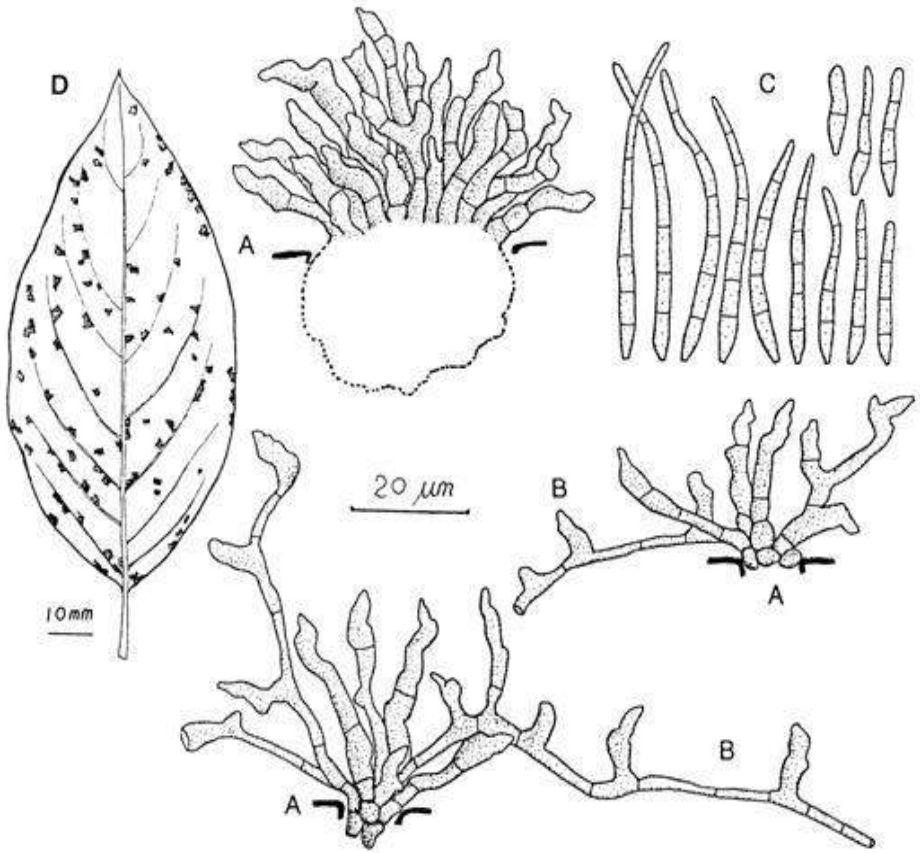
- Spots numerous , enlarge - **entire leaflet withers and dies**
- On petioles, stems- elliptical, brownish lesions with pale center
- **Spots are darker- in humid weather**
- Lower surface of the **lesion appears light gray or silvery due to hyaline conidia**

Carrot - Cercospora leaf spot *Cercospora carotae, C.apii*

Symtoms contd…

- Lesions - merge and girdle -collapse and death of the leaf /stem
- **Tops break easily- mechanical harvesting, leaving the roots in the ground.**
- Floral parts- if infected early - **shrivelled seed**
- Later infection – **cause seed infection- seed-borne inoculum.**
- **Survives in or on seed, crop residue, weed hosts, wild carrot**

Cercospora spp

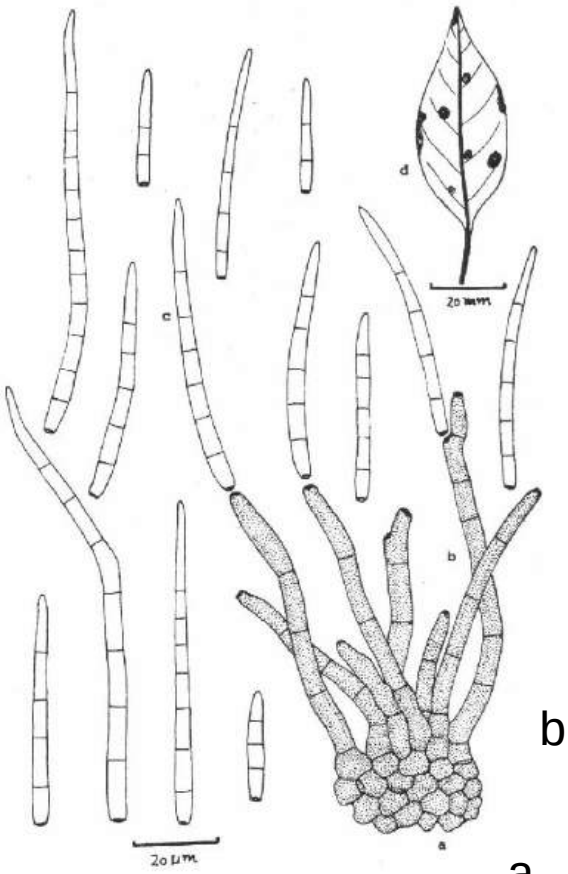


Conidiophore shows distinct geniculate bends and scar after release of conidia

Conidiophores may be continuous or septate, straight or flexuous, geniculate, denticulate, hyaline or of various shades and tints of brown up to blackish brown.

The growth of the conidiophores can be determinate or sympodial.

Cercospora spp



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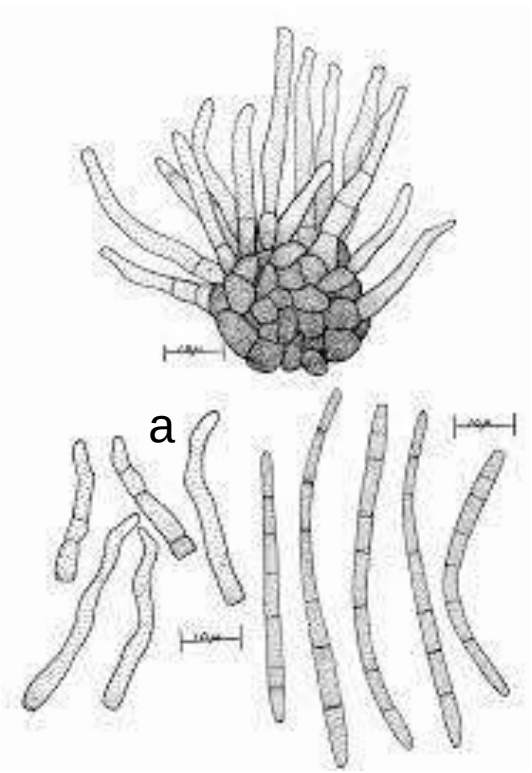

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a. Stroma, b. Conidiophore; c. Conidium,

Cercospora spp

- **Conidia :** long, slender, cylindrical, straight to slightly curved, multiseptate, hyaline or brownish conidia on short dark conidiophores.
- The conidia vary in size but always have a breadth, length ratio of 1:10-1:150
- One of the most important features of conidial morphology is the presence a conspicuous, thickened and dark scar called hilum, which indicates the point of attachment to the conidiophores.

Carrot - Cercospora leaf spot *Cercospora carotae, C.apii*

Management:

- **Seed production in disease free areas**
- Crop rotation for 2-3 years
- Tolerant var- Delite, Delux, Fancy, Bonus, Classic, Winner and Premium
- Sanitation
- Seed treatment – soaking in Thiram for 24 hrs
- Mancozeb (2%) - 3-4 sprays from disease appearance
- **Difenconazole 125g/ha at 14 days interval**

Alternaria leaf blight - *Alternaria dauci, A. radicina*

Dark brown to black lesions with a yellow halo along the margins

Leaflets shrivel and die, with blighted appearance

Alternaria leaf blight - *Alternaria dauci, A. radicina*

On petiole- lesions merge - girdle entire leaf - collapse and die

Blackening and shrivelling progress rapidly

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Blighted carrot tops may break off

Alternaria leaf blight - *Alternaria dauci, A. radicina*

- ***A. dauci* is most common** and destroys the foliage (Tops)
- Carrot Tops essential - mechanical harvesting.

Symptoms:

- Early seedling infection - **damping-off.**
- Lesions on plants **similar to cercospora leaf spot but more irregular in shape**
- Alternaria blight often **appears later than cercospora blight**
- Lesions –dark brown to black with a yellow halo, along the margins
- Leaflets shrivel and die, with blighted appearance

Alternaria leaf blight - *Alternaria dauci, A. radicina*

Symptoms contd…

- Under cool, humid conditions**, a velvety mycelial growth** - visible to the naked eye.
- On petiole - lesions merge - girdle entire leaf - collapse and die
- Blackening and shrivelling progress rapidly – **like frost injury in moist weather**
- **Blighted carrot tops may break off leaving the roots in the ground** when gripped by mechanical harvesters,
- On **fleshy roots – some times black decay**

Alternaria spp - Etiology

The mycelium is branched, septate and inter (First) and intra cellular and pale to olivaceous brown or dark brown.

Conidiophores: arise singly or in groups (fascicles), through stomata from the mycelium present in the dead centres of the spots.

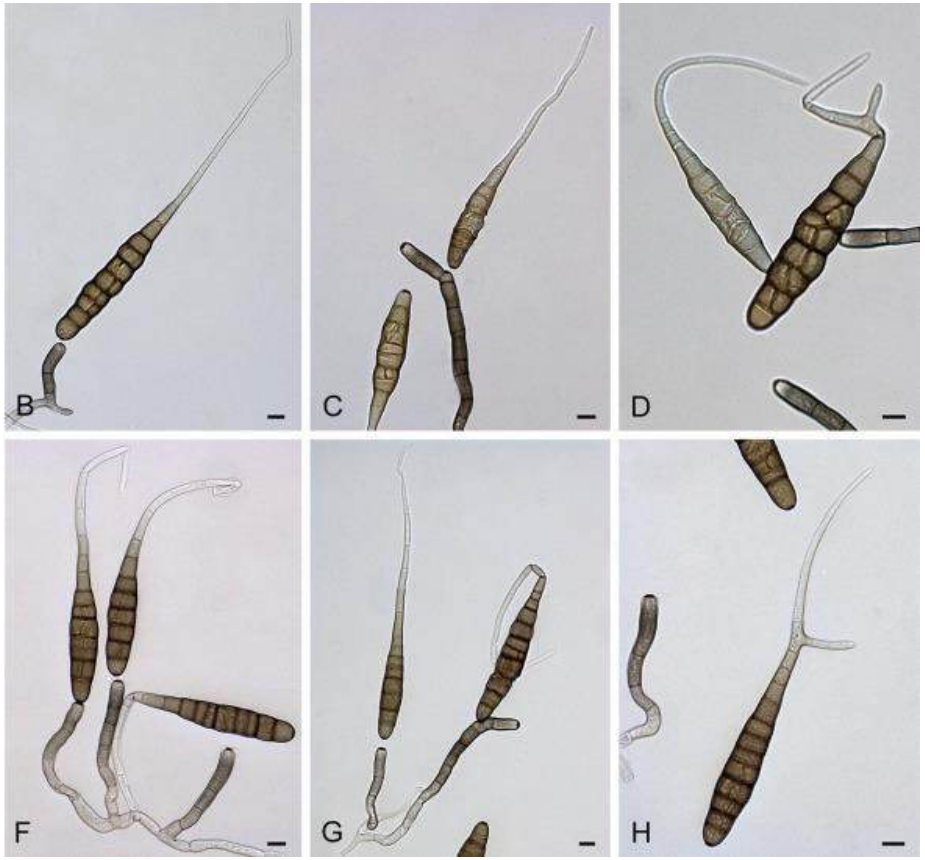
Conidiophores - simple, straight or curved , 1-3 septate , dark coloured.

The formation of polymorphous dark colour conidia either singly or in short or longer chains (Catenulate).

Conidia Muriform shaped, multi-celled, brown, obclavate with a beak, transversely septate and 1-2 longitudinally or obliquely septate, Conidia are produced acropetally (new one at the top) in chains (catenulate) through the pores formed at the apex of the beak of conidia.

Ascospores are produced in bi-tunicate asci (ascostoma)

Alternaria spp



Alternaria leaf blight - *Alternaria dauci, A. radicina*

Survives in or on seed, crop residue, weed hosts

Management:

- **Cultural practices** : Field sanitation, crop rotation
- Use of disease free seed
- Deep ploughing
- **Hot water treat. 50°C/25 minutes**
- Resistant lines- **IIHR-230,231,232,248, 253 and 264**
- Seed treatment with thiram/captan 3g/kg

Alternaria leaf blight - *Alternaria dauci, A. radicina*

Management contd…

- **Biopriming** the seed with ***Gliocladium roseum* (*Syn.Clonostachys rosea*)** strain IK 726
- **Application of N fertilizers**
- Urea as foliar spray at the end of the season **- new foliage, helps in mechanical harvesting**
- Monitoring
- Start sprays **25% of plants (field) with - middle diseased leaf** - accompanied by a rain forecast or a minimum temperature >16°C - next night.
- **If conditions persist, subsequent sprays at 7 to 10 days interval.**

Powdery mildew - *Erysiphe polygoni, E. heraclei*

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More serious in **winter sown crop- planted in January**

Disease appears **- June- July -before flowering**

In late sown crop **seed yield loss up to 80%**

Symptoms :

- **Younger and inner leaves** are affected initially – soon coalesce covering large areas
- Cob web mycelial growth on upper leaf surface
- Younger plants –vigour greatly reduced/ killed
- Slight necrosis when slightly affected

Powdery mildew - *Erysiphe polygoni, E. heraclei*

- Wide host range – 85 umbelliferous genera- celery, fennel, coriander, dill, parsely, parsnip
- Either as cleistothecia/conidia in the perennial hosts
- **Seed transmission**
- **Perfect stage** has been reported on some cultivars at IIHR
- **Management:**
- Rest lines- **IIHR-230,231,232,248,253 and 264**
- Wettable sulphur, karathane, carbendazim, hexaconazole

Bacterial blight - *Xanthomonas campestris* pv *carotae*

- On leaves- Lesions irregular, dark brown, necrotic in the center
- Later - dry and brittle with yellow halo

Bacterial blight - *Xanthomonas campestris* pv *carotae*

Infection - often along the leaf margin, hence lateral curling

Bacterial blight - *Xanthomonas campestris* pv *carotae*

Fig. 1. Umbel blight of carrot caused by *Xanthomonas campestris* pv. *carotae*.

Lesions on the petioles, peduncles and pedicels appear water-soaked and are linear

Young flower umbel -If attacked - entire umbel -killed or blighted

Bacterial blight - *Xanthomonas campestris* pv *carotae*

Symptoms

- On leaves- Lesions irregular, dark brown, necrotic in the center
- Later - dry and brittle with yellow halo
- In contrast, lesions on the petioles, peduncles and pedicels appear water-soaked and are linear.
- Infection - often along the leaf margin, hence lateral curling
- Young flower umbel -If attacked -**entire umbel -killed or blighted.**

Bacterial blight - *Xanthomonas campestris* pv *carotae*

Symptoms contd…

- If infection -after umbel formation, only part of it may be affected.
- A gummy bacterial exudate sometimes collects and flows down from the infected plant parts
- Disease- severe, most of the leaflets turn yellow and lower leaves die
- **Early symptoms can be mistaken for Cercospora/ Alternaria blights – difference is presence of bacterial exudate**

Bacterial blight - *Xanthomonas campestris* pv *carotae*

Survives **in or on seed** , also persist in soil on crop residues for one winter season.

Management

- Seed production - disease free areas
- 2-3 year rotation
- **Seed treatment - hot water at 52°C for 25 minutes**
- **Tolerant varieties - Waltham Hi-Color, Danvers**
- Removal and destruction of crop residues after harvest
- Spray twice with Streptocycline 1.0 g mixed with copper oxychloride 30 g per 10 litre of water at 15 days interval

Root scab - *Streptomyces scabies*

Scab symptoms on the roots - corky tissue darker than healthy tissue

Lesions may be sunken/raised on surface of the skin

Root scab - *Streptomyces scabies*

Other crops-Beet, Potato, turnip, radish and parsnip

- **Disease more severe in dry, alkaline soils.**

Symptoms:

- ***S. scabies* cause damping-off**
- Scab symptoms on the roots
- Abnormal growth of the host cells- corky tissue darker than healthy tissue.
- Lesions may be sunken/raised on surface of the skin
- Lesions merge with more scabby area

Root scab - *Streptomyces scabies*

Management

- Apply - **fertilizers ammonium sulphate or sulphur, to lower the soil pH.**
- Maintain proper soil moisture / growing in soils with more holding capacity
- **Carrot should not be cultivated in potato fields**
- Long rotations with small grains, grasses or corn

Bacterial soft rot - *Erwinia carotovora* pv. *Carotovora*, *E. carotovora pv. atroseptica*

Rotted roots become watery and mushy

Bacterial soft rot *Erwinia carotovora* pv. *Carotovora*, *E. carotovora pv. atroseptica*

The surface becomes depressed , lesions turn dark

Bacterial soft rot *Erwinia carotovora* pv. *Carotovora*, *E. carotovora pv. atroseptica*

- In the field, soft rot – pits on root, with a distinct margin between diseased and healthy tissue.

- **The lesions soon become opaque and slimy.**

Bacterial soft rot- Weak parasite

Erwinia carotovora pv. Carotovora, E. carotovora pv. *atroseptica*

- Common disease- **major losses in storage under natural conditions- as a secondary invader**
- **Not imp.- In cold stored carrots**
- Sporadically -major losses - up to 70% - field
- **Field infections -associated with extreme wet conditions**

Symptoms:

- On **lower part** of roots - small, water soaked lesions, enlarge and coalesce.
- Spread rapidly **to the core of the root and progress to outside**

Bacterial soft rot- Weak parasite

Erwinia carotovora pv. Carotovora, E. carotovora pv. *atroseptica*

Symptoms contd…

- Soon rot, **become watery and mushy,**
- The surface becomes depressed , lesions turn dark
- The outer surface often remains intact over the liquified interior, but is easily broken if the carrots are moved.
- The lesions soon become **opaque and slimy**.
- **Cracks often - on root surface -macerated tissue oozed out.**

Bacterial soft rot- Weak parasite

Erwinia carotovora pv. Carotovora, E. carotovora pv. *atroseptica*

Symptoms contd…

- When it is exposed to the air, it turns tan or gray.
- Plants exhibit wilting symptoms
- **Foul odor - bacterial soft rot** until the tissues collapse and secondary organisms invade.
- ***Sclerotinia sclerotiorum*** - also cause a soft rot of carrot
  - with the presence of white mycelium
- **Infection by Sclerotinia and Rhizoctonia fungi can predispose carrots to attack by soft rot bacteria**

Bacterial soft rot- Weak parasite

Erwinia carotovora pv. Carotovora, E. carotovora pv. *atroseptica*

Symptoms contd…

- In the field, soft rot – pits on root, with a distinct margin between diseased and healthy tissue.
- Under **waterlogged conditions-** rotted roots may remain in the ground- when plant is pulled.
- **If drier soil conditions return, root continue to grow, but stunted and hence unmarketable**

Bacterial soft rot- Weak parasite

Erwinia carotovora pv. Carotovora, E. carotovora pv. *atroseptica*

Seed tubers, weeds , plant debris as inoculum

Management

- **Grow carrot in well-drained soil**
- **Rola cv. Least susceptible**
- Seed production - disease free areas
- **Crop rotation for 2-3years, corn, small grains, grasses, alfalfa, clover, beet and bean**
- Avoid bruises during harvesting and handling

Bacterial soft rot- Weak parasite

Erwinia carotovora pv. Carotovora, E. carotovora pv. *atroseptica*

Management contd…

- **Wash the tubers with chlorinated water, dry properly– before storing**
- Sort out the damaged and diseased roots
- **Store carrots close to 0°C and 90 to 95% relative humidity**
- **Disinfection of stores**
- Proper aeration during storage

Carrot Yellows /Aster yellows- Phytoplasma

Aster Yellows symptoms on carrot includes yellowing of foliage.
Courtesy J.A. Amador, TAES, Weslaco, 1996.

Leaf yellowing, with some vein clearing of the younger leaves at the center of the crown

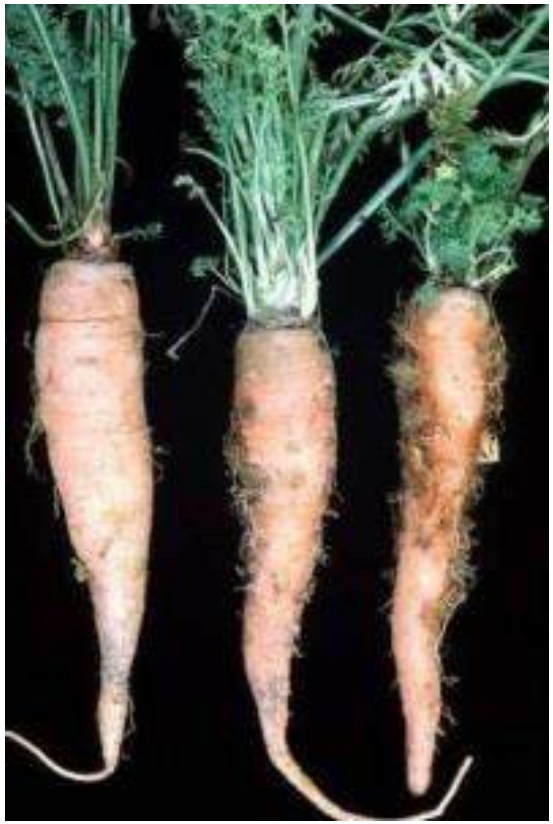
Carrot Yellows /Aster yellows- Phytoplasma

Older leaves are **whitish at first, then become bronze, reddened or both.**

Carrot Yellows /Aster yellows- Phytoplasma

- **Infected plants produce numerous filamentous stems and simple leaves**
- **Plants bushy - broom like, small, chlorotic leaves**

Carrot Yellows /Aster yellows- Phytoplasma



Many malformed fibrous rootlets usually appear in rows along the vertical axis of the main root .

Carrot Yellows /Aster yellows- Phytoplasma

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Carrot Yellows /Aster yellows- Phytoplasma

- **Flowers are small, malformed, and often remain green or fail to develop the proper colour.**
- **Sterility of the flowering umbels**

Carrot Yellows /Aster yellows- Phytoplasma

Minor disease - most common in the traditional areas
Prevalence varies from year to **year- generally <than 2% of carrot area**

Symptoms:

- Leaf yellowing, with some vein clearing of the younger leaves at the center of the crown
- Later a mass new shoots grows from the crown- **sickly - witches' broom appearance**
- Older leaves are whitish at first, then become bronze, reddened or both.

Carrot Yellows /Aster yellows- Phytoplasma

Symptoms contd…

- The **reddened leaves are distinctive -recognizable in the field**
- The **petioles are twisted and eventually break off leaving a short bunchy top**
- Not suitable for mechanical harvesting / bunching- fresh market.
- Many malformed **fibrous rootlets** usually appear in rows along the vertical axis of the main root .
- **The color, texture, flavor of roots altered with bitter and astringent taste.**

Carrot Yellows /Aster yellows- Phytoplasma

Symptoms contd…

- The crowns affected by **bacterial soft rot under** wet weather.
- The disease can continue to develop in storage.
- On seed plants, various degrees of stunting occur, as well as malformation, chlorosis and **sterility of the flowering umbels.**
- The severity of disease and damage varies with crop age
- Most severe on **late-harvested crops.**
- Wide host range than includes many vegetable crops.

Carrot Yellows /Aster yellows- Phytoplasma

Survival and spread

Vector: Six-spotted leaf hopper -*Macrosteles divisus, Macrosteles quadripunctulatus Macrosteles sexnotatus Macrosteles laevis*

Over 350 plants including many common vegetables, annual flowering plants, perennial flowering plants and weeds.

Carrot Yellows /Aster yellows- Phytoplasma

Management:

- Avoid planting carrot near fields of lettuce or other susceptible crops.
- Destruction of crop debris immediately after harvest.
- Spaying of systemic insecticides - **Methyl demeton, 2 ml/l or Dimethoate 2 ml/l or Imidachloprid 1.2 ml/l**